



MATH240: Applications Growth Decay

Introduction to Statistics · C-ID MATH240 · numberbender.com

Name: _____

Date: _____

Score: / 30

Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Exponential growth and decay

1. Newton's Law of Cooling: $T'(t) = -4(T(t) - 20)$, where 20 is the ambient temperature. If the object is currently $T = 20 + 27$ degrees above ambient (so $T = 20 + 27$), find $T'(t)$.

$$f(x) = -4x + 20, \quad x = 27$$

Answer: _____

2. A population satisfies $y' = 1y$. If the initial population is $y(0) = 46$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 1x + 46, \quad x = 0$$

Answer: _____

3. A radioactive substance decays. The amount $Q(t) = -2t^2 + 183$ grams at time t seconds. Find $Q(2)$.

$$-2x^2 + 183, \quad x = 2$$

Answer: _____

4. A tank contains Q grams of salt. The rate of change is $Q'(t) = -2Q + 8$. Find $Q'(t)$ when $Q = 20$ grams.

$$f(x) = -2x + 8, \quad x = 20$$

Answer: _____

5. Newton's Law of Cooling: $T'(t) = -4(T(t) - 23)$, where 23 is the ambient temperature. If the object is currently $T = 23 + 12$ degrees above ambient (so $T = 23 + 12$), find $T'(t)$.

$$f(x) = -4x + 23, \quad x = 12$$

Answer: _____



6. A population satisfies $y' = 3y$. If the initial population is $y(0) = 18$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 3x + 18, \quad x = 0$$

Answer: _____

7. A radioactive substance decays. The amount $Q(t) = -1t^2 + 130$ grams at time t seconds. Find $Q(3)$.

$$-1x^2 + 130, \quad x = 3$$

Answer: _____

8. A tank contains Q grams of salt. The rate of change is $Q'(t) = -3Q + 7$. Find $Q'(t)$ when $Q = 18$ grams.

$$f(x) = -3x + 7, \quad x = 18$$

Answer: _____

9. Newton's Law of Cooling: $T'(t) = -1(T(t) - 22)$, where 22 is the ambient temperature. If the object is currently $T = 22 + 33$ degrees above ambient (so $T = 22 + 33$), find $T'(t)$.

$$f(x) = -1x + 22, \quad x = 33$$

Answer: _____

10. A population satisfies $y' = 2y$. If the initial population is $y(0) = 74$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 2x + 74, \quad x = 0$$

Answer: _____

11. A radioactive substance decays. The amount $Q(t) = -3t^2 + 63$ grams at time t seconds. Find $Q(5)$.

$$-3x^2 + 63, \quad x = 5$$

Answer: _____

12. A tank contains Q grams of salt. The rate of change is $Q'(t) = -1Q + 15$. Find $Q'(t)$ when $Q = 11$ grams.

$$f(x) = -1x + 15, \quad x = 11$$

Answer: _____

13. Newton's Law of Cooling: $T'(t) = -3(T(t) - 23)$, where 23 is the ambient temperature. If the object is currently $T = 23 + 45$ degrees above ambient (so $T = 23 + 45$), find $T'(t)$.

$$f(x) = -3x + 23, \quad x = 45$$

Answer: _____



14. A population satisfies $y' = 1y$. If the initial population is $y(0) = 35$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 1x + 35, \quad x = 0$$

Answer: _____

15. A radioactive substance decays. The amount $Q(t) = -1t^2 + 52$ grams at time t seconds. Find $Q(5)$.

$$-1x^2 + 52, \quad x = 5$$

Answer: _____

16. A tank contains Q grams of salt. The rate of change is $Q'(t) = -3Q + 7$. Find $Q'(t)$ when $Q = 5$ grams.

$$f(x) = -3x + 7, \quad x = 5$$

Answer: _____

17. Newton's Law of Cooling: $T'(t) = -3(T(t) - 21)$, where 21 is the ambient temperature. If the object is currently $T = 21 + 22$ degrees above ambient (so $T = 21 + 22$), find $T'(t)$.

$$f(x) = -3x + 21, \quad x = 22$$

Answer: _____

18. A population satisfies $y' = 2y$. If the initial population is $y(0) = 20$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 2x + 20, \quad x = 0$$

Answer: _____

19. A radioactive substance decays. The amount $Q(t) = -3t^2 + 122$ grams at time t seconds. Find $Q(3)$.

$$-3x^2 + 122, \quad x = 3$$

Answer: _____

20. A tank contains Q grams of salt. The rate of change is $Q'(t) = -2Q + 4$. Find $Q'(t)$ when $Q = 11$ grams.

$$f(x) = -2x + 4, \quad x = 11$$

Answer: _____

21. Newton's Law of Cooling: $T'(t) = -3(T(t) - 20)$, where 20 is the ambient temperature. If the object is currently $T = 20 + 25$ degrees above ambient (so $T = 20 + 25$), find $T'(t)$.

$$f(x) = -3x + 20, \quad x = 25$$

Answer: _____



22. A population satisfies $y' = 4y$. If the initial population is $y(0) = 66$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 4x + 66, \quad x = 0$$

Answer: _____

23. A radioactive substance decays. The amount $Q(t) = -2t^2 + 81$ grams at time t seconds. Find $Q(5)$.

$$-2x^2 + 81, \quad x = 5$$

Answer: _____

24. A tank contains Q grams of salt. The rate of change is $Q'(t) = -2Q + 6$. Find $Q'(t)$ when $Q = 14$ grams.

$$f(x) = -2x + 6, \quad x = 14$$

Answer: _____

25. Newton's Law of Cooling: $T'(t) = -4(T(t) - 22)$, where 22 is the ambient temperature. If the object is currently $T = 22 + 37$ degrees above ambient (so $T = 22 + 37$), find $T'(t)$.

$$f(x) = -4x + 22, \quad x = 37$$

Answer: _____

26. A population satisfies $y' = 1y$. If the initial population is $y(0) = 24$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 1x + 24, \quad x = 0$$

Answer: _____

27. A radioactive substance decays. The amount $Q(t) = -2t^2 + 178$ grams at time t seconds. Find $Q(2)$.

$$-2x^2 + 178, \quad x = 2$$

Answer: _____

28. A tank contains Q grams of salt. The rate of change is $Q'(t) = -1Q + 2$. Find $Q'(t)$ when $Q = 8$ grams.

$$f(x) = -1x + 2, \quad x = 8$$

Answer: _____

29. Newton's Law of Cooling: $T'(t) = -4(T(t) - 22)$, where 22 is the ambient temperature. If the object is currently $T = 22 + 39$ degrees above ambient (so $T = 22 + 39$), find $T'(t)$.

$$f(x) = -4x + 22, \quad x = 39$$

Answer: _____



30. A population satisfies $y' = 3y$. If the initial population is $y(0) = 75$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 3x + 75, \quad x = 0$$

Answer: _____





MATH240: Applications Growth Decay

Introduction to Statistics · C-ID MATH240 · numberbender.com

ANSWER KEY & SOLUTIONS

Topics: Exponential growth and decay. All answers verified by independent computation.

Solutions



Exponential growth and decay

1. Newton's Law of Cooling: $T'(t) = -4(T(t) - 20)$, where 20 is the ambient temperature. If the object is currently $T = 20 + 27$ degrees above ambient (so $T = 20 + 27$), find $T'(t)$.

$$f(x) = -4x + 20, \quad x = 27$$

$$\rightarrow T'(t) = -4 \cdot (T - 20) = -4 \cdot 27 = -88 \text{ degrees/sec.}$$

$\rightarrow$ Negative rate means the object is cooling toward ambient.

Answer: $f(27) = -88$

2. A population satisfies $y' = 1y$. If the initial population is $y(0) = 46$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 1x + 46, \quad x = 0$$

$$\rightarrow y'(t) = 1y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 1 \cdot y(0) = 1 \cdot 46 = 46.$$

Answer: $f(0) = 46$

3. A radioactive substance decays. The amount $Q(t) = -2t^2 + 183$ grams at time t seconds. Find $Q(2)$.

$$-2x^2 + 183, \quad x = 2$$

$$\rightarrow Q(2) = -2(2)^2 + 183 = -8 + 183 = 175 \text{ grams.}$$

Answer: $-2(2)^2 + 183 = -8 + 183 = 175$

4. A tank contains Q grams of salt. The rate of change is $Q'(t) = -2Q + 8$. Find $Q'(t)$ when $Q = 20$ grams.

$$f(x) = -2x + 8, \quad x = 20$$

$$\rightarrow \text{Substitute } Q = 20: Q'(20) = -2 \cdot 20 + 8 = -40 + 8 = -32 \text{ g/min.}$$

Answer: $f(20) = -32$

5. Newton's Law of Cooling: $T'(t) = -4(T(t) - 23)$, where 23 is the ambient temperature. If the object is currently $T = 23 + 12$ degrees above ambient (so $T = 23 + 12$), find $T'(t)$.

$$f(x) = -4x + 23, \quad x = 12$$

$$\rightarrow T'(t) = -4 \cdot (T - 23) = -4 \cdot 12 = -25 \text{ degrees/sec.}$$

$\rightarrow$ Negative rate means the object is cooling toward ambient.

Answer: $f(12) = -25$

6. A population satisfies $y' = 3y$. If the initial population is $y(0) = 18$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 3x + 18, \quad x = 0$$

$$\rightarrow y'(t) = 3y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 3 \cdot y(0) = 3 \cdot 18 = 18.$$

Answer: $f(0) = 18$



7. A radioactive substance decays. The amount $Q(t) = -1t^2 + 130$ grams at time t seconds. Find $Q(3)$.

$$-1x^2 + 130, \quad x = 3$$

$$\rightarrow Q(3) = -1(3)^2 + 130 = -9 + 130 = 121 \text{ grams.}$$

Answer: $-1(3)^2 + 130 = -9 + 130 = 121$

8. A tank contains Q grams of salt. The rate of change is $Q'(t) = -3Q + 7$. Find $Q'(t)$ when $Q = 18$ grams.

$$f(x) = -3x + 7, \quad x = 18$$

$$\rightarrow \text{Substitute } Q = 18: Q'(18) = -3 \cdot 18 + 7 = -54 + 7 = -47 \text{ g/min.}$$

Answer: $f(18) = -47$

9. Newton's Law of Cooling: $T'(t) = -1(T(t) - 22)$, where 22 is the ambient temperature. If the object is currently $T = 22 + 33$ degrees above ambient (so $T = 22 + 33$), find $T'(t)$.

$$f(x) = -1x + 22, \quad x = 33$$

$$\rightarrow T'(t) = -1 \cdot (T - 22) = -1 \cdot 33 = -11 \text{ degrees/sec.}$$

$$\rightarrow \text{Negative rate means the object is cooling toward ambient.}$$

Answer: $f(33) = -11$

10. A population satisfies $y' = 2y$. If the initial population is $y(0) = 74$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 2x + 74, \quad x = 0$$

$$\rightarrow y'(t) = 2y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 2 \cdot y(0) = 2 \cdot 74 = 74.$$

Answer: $f(0) = 74$

11. A radioactive substance decays. The amount $Q(t) = -3t^2 + 63$ grams at time t seconds. Find $Q(5)$.

$$-3x^2 + 63, \quad x = 5$$

$$\rightarrow Q(5) = -3(5)^2 + 63 = -75 + 63 = -12 \text{ grams.}$$

Answer: $-3(5)^2 + 63 = -75 + 63 = -12$

12. A tank contains Q grams of salt. The rate of change is $Q'(t) = -1Q + 15$. Find $Q'(t)$ when $Q = 11$ grams.

$$f(x) = -1x + 15, \quad x = 11$$

$$\rightarrow \text{Substitute } Q = 11: Q'(11) = -1 \cdot 11 + 15 = -11 + 15 = 4 \text{ g/min.}$$

Answer: $f(11) = 4$

13. Newton's Law of Cooling: $T'(t) = -3(T(t) - 23)$, where 23 is the ambient temperature. If the object is currently $T = 23 + 45$ degrees above ambient (so $T = 23 + 45$), find $T'(t)$.

$$f(x) = -3x + 23, \quad x = 45$$

$$\rightarrow T'(t) = -3 \cdot (T - 23) = -3 \cdot 45 = -112 \text{ degrees/sec.}$$

$$\rightarrow \text{Negative rate means the object is cooling toward ambient.}$$

Answer: $f(45) = -112$



14. A population satisfies $y' = 1y$. If the initial population is $y(0) = 35$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 1x + 35, \quad x = 0$$

$$\rightarrow y'(t) = 1y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 1 \cdot y(0) = 1 \cdot 35 = 35.$$

Answer: $f(0) = 35$

15. A radioactive substance decays. The amount $Q(t) = -1t^2 + 52$ grams at time t seconds. Find $Q(5)$.

$$-1x^2 + 52, \quad x = 5$$

$$\rightarrow Q(5) = -1(5)^2 + 52 = -25 + 52 = 27 \text{ grams.}$$

Answer: $-1(5)^2 + 52 = -25 + 52 = 27$

16. A tank contains Q grams of salt. The rate of change is $Q'(t) = -3Q + 7$. Find $Q'(t)$ when $Q = 5$ grams.

$$f(x) = -3x + 7, \quad x = 5$$

$$\rightarrow \text{Substitute } Q = 5: Q'(5) = -3 \cdot 5 + 7 = -15 + 7 = -8 \text{ g/min.}$$

Answer: $f(5) = -8$

17. Newton's Law of Cooling: $T'(t) = -3(T(t) - 21)$, where 21 is the ambient temperature. If the object is currently $T = 21 + 22$ degrees above ambient (so $T = 21 + 22$), find $T'(t)$.

$$f(x) = -3x + 21, \quad x = 22$$

$$\rightarrow T'(t) = -3 \cdot (T - 21) = -3 \cdot 22 = -45 \text{ degrees/sec.}$$

$$\rightarrow \text{Negative rate means the object is cooling toward ambient.}$$

Answer: $f(22) = -45$

18. A population satisfies $y' = 2y$. If the initial population is $y(0) = 20$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 2x + 20, \quad x = 0$$

$$\rightarrow y'(t) = 2y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 2 \cdot y(0) = 2 \cdot 20 = 20.$$

Answer: $f(0) = 20$

19. A radioactive substance decays. The amount $Q(t) = -3t^2 + 122$ grams at time t seconds. Find $Q(3)$.

$$-3x^2 + 122, \quad x = 3$$

$$\rightarrow Q(3) = -3(3)^2 + 122 = -27 + 122 = 95 \text{ grams.}$$

Answer: $-3(3)^2 + 122 = -27 + 122 = 95$

20. A tank contains Q grams of salt. The rate of change is $Q'(t) = -2Q + 4$. Find $Q'(t)$ when $Q = 11$ grams.

$$f(x) = -2x + 4, \quad x = 11$$

$$\rightarrow \text{Substitute } Q = 11: Q'(11) = -2 \cdot 11 + 4 = -22 + 4 = -18 \text{ g/min.}$$

Answer: $f(11) = -18$



21. Newton's Law of Cooling: $T'(t) = -3(T(t) - 20)$, where 20 is the ambient temperature. If the object is currently $T = 20 + 25$ degrees above ambient (so $T = 20 + 25$), find $T'(t)$.

$$f(x) = -3x + 20, \quad x = 25$$

$$\rightarrow T'(t) = -3 \cdot (T - 20) = -3 \cdot 25 = -55 \text{ degrees/sec.}$$

$\rightarrow$ Negative rate means the object is cooling toward ambient.

Answer: $f(25) = -55$

22. A population satisfies $y' = 4y$. If the initial population is $y(0) = 66$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 4x + 66, \quad x = 0$$

$$\rightarrow y'(t) = 4y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 4 \cdot y(0) = 4 \cdot 66 = 66.$$

Answer: $f(0) = 66$

23. A radioactive substance decays. The amount $Q(t) = -2t^2 + 81$ grams at time t seconds. Find $Q(5)$.

$$-2x^2 + 81, \quad x = 5$$

$$\rightarrow Q(5) = -2(5)^2 + 81 = -50 + 81 = 31 \text{ grams.}$$

Answer: $-2(5)^2 + 81 = -50 + 81 = 31$

24. A tank contains Q grams of salt. The rate of change is $Q'(t) = -2Q + 6$. Find $Q'(t)$ when $Q = 14$ grams.

$$f(x) = -2x + 6, \quad x = 14$$

$$\rightarrow \text{Substitute } Q = 14: Q'(14) = -2 \cdot 14 + 6 = -28 + 6 = -22 \text{ g/min.}$$

Answer: $f(14) = -22$

25. Newton's Law of Cooling: $T'(t) = -4(T(t) - 22)$, where 22 is the ambient temperature. If the object is currently $T = 22 + 37$ degrees above ambient (so $T = 22 + 37$), find $T'(t)$.

$$f(x) = -4x + 22, \quad x = 37$$

$$\rightarrow T'(t) = -4 \cdot (T - 22) = -4 \cdot 37 = -126 \text{ degrees/sec.}$$

$\rightarrow$ Negative rate means the object is cooling toward ambient.

Answer: $f(37) = -126$

26. A population satisfies $y' = 1y$. If the initial population is $y(0) = 24$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 1x + 24, \quad x = 0$$

$$\rightarrow y'(t) = 1y(t).$$

$$\rightarrow \text{At } t=0: y'(0) = 1 \cdot y(0) = 1 \cdot 24 = 24.$$

Answer: $f(0) = 24$

27. A radioactive substance decays. The amount $Q(t) = -2t^2 + 178$ grams at time t seconds. Find $Q(2)$.

$$-2x^2 + 178, \quad x = 2$$

$$\rightarrow Q(2) = -2(2)^2 + 178 = -8 + 178 = 170 \text{ grams.}$$

Answer: $-2(2)^2 + 178 = -8 + 178 = 170$



28. A tank contains Q grams of salt. The rate of change is $Q'(t) = -1Q + 2$. Find $Q'(t)$ when $Q = 8$ grams.

$$f(x) = -1x + 2, \quad x = 8$$

→ Substitute $Q = 8$: $Q'(8) = -1 \cdot 8 + 2 = -8 + 2 = -6$ g/min.

Answer: $f(8) = -6$

29. Newton's Law of Cooling: $T'(t) = -4(T(t) - 22)$, where 22 is the ambient temperature. If the object is currently $T = 22 + 39$ degrees above ambient (so $T = 22 + 39$), find $T'(t)$.

$$f(x) = -4x + 22, \quad x = 39$$

→ $T'(t) = -4 \cdot (T - 22) = -4 \cdot 39 = -134$ degrees/sec.

→ Negative rate means the object is cooling toward ambient.

Answer: $f(39) = -134$

30. A population satisfies $y' = 3y$. If the initial population is $y(0) = 75$, find the initial rate of growth dy/dt at $t = 0$.

$$f(x) = 3x + 75, \quad x = 0$$

→ $y'(t) = 3y(t)$.

→ At $t=0$: $y'(0) = 3 \cdot y(0) = 3 \cdot 75 = 75$.

Answer: $f(0) = 75$

